

```
import matplotlib.pyplot as plt

plt.figure(figsize=(10,1))
plt.text(0.1,0.5,"🔴 Razorpay AI Fraud Risk Dashboard",
        fontsize=22, weight="bold")
plt.axis("off")
plt.show()
```

/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151: UserWarning:
Glyph 128179 (\N{CREDIT CARD}) missing from font(s) DejaVu Sans.

🔴 Razorpay AI Fraud Risk Dashboard

```
import pandas as pd
import matplotlib.pyplot as plt

results = pd.read_csv("/content/Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")
results.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...	V27	V28	Hou
0	51014	1.468115	-1.230332	0.740993	-1.283638	-1.976458	-0.851640	-1.315321	-0.073429	-1.387000	...	0.038051	0.025579	1
1	56749	1.246281	-0.719496	1.368630	0.473247	-1.133748	1.141729	-1.408090	0.343605	0.120501	...	0.146494	0.038369	1
2	39487	1.175250	0.207760	0.748263	1.126846	-0.597513	-0.759688	-0.065497	-0.102567	0.213570	...	0.032508	0.037459	1
3	42819	0.916167	-0.922102	0.643763	0.070063	-0.878645	0.467869	-0.541612	0.196626	0.945984	...	-0.067281	0.024257	1
4	60140	-1.052824	0.842871	1.097023	0.114082	0.083399	1.634920	-0.399411	0.999016	-0.832466	...	-0.264866	-0.012381	1

5 rows × 37 columns

```
print("🟡 Total Transactions:", len(results))
print("🟢 Approved:", (results["Action"]=="Approve").sum())
print("🟡 Manual Review:", (results["Action"]=="Manual Review").sum())
print("🔴 Blocked:", (results["Action"]=="Block Transaction").sum())
print("📊 Average Risk Score:", round(results["Risk_Score"].mean(),2))
```

```
results["Action"].value_counts().plot(kind="bar")
plt.title("AI Payment Decision Summary")
plt.xlabel("Decision")
plt.ylabel("Number of Transactions")
plt.grid(axis="y")
plt.show()
```

```
results["Action"].value_counts().plot(kind="bar")
plt.title("AI Payment Decision Summary")
plt.xlabel("Decision")
plt.ylabel("Transactions")
plt.grid(True)
plt.show()
```

```
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151: UserWarning:
```

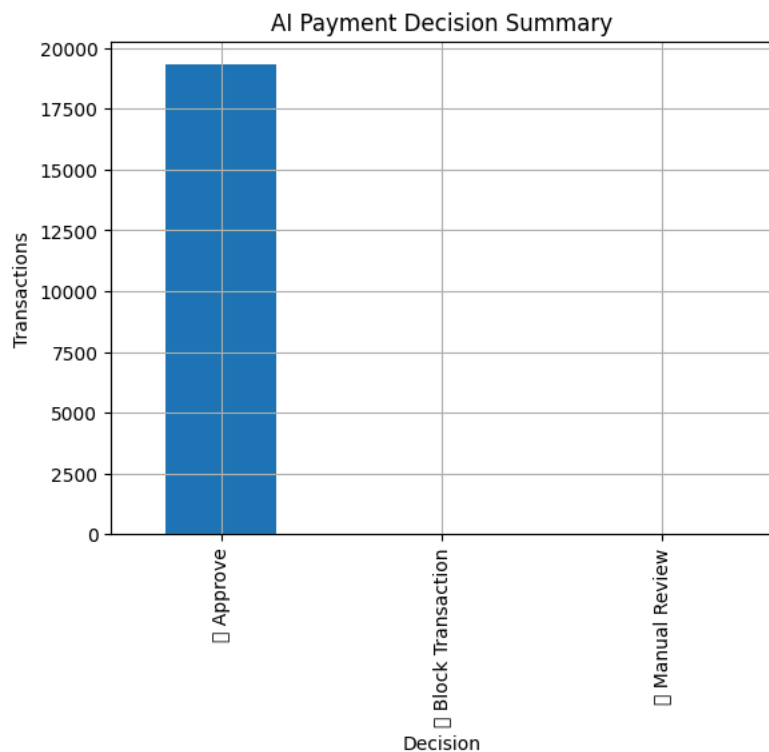
```
Glyph 128994 (\N{LARGE GREEN CIRCLE}) missing from font(s) DejaVu Sans.
```

```
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151: UserWarning:
```

```
Glyph 128683 (\N{NO ENTRY SIGN}) missing from font(s) DejaVu Sans.
```

```
/usr/local/lib/python3.13/dist-packages/IPython/core/pylabtools.py:151: UserWarning:
```

```
Glyph 128993 (\N{LARGE YELLOW CIRCLE}) missing from font(s) DejaVu Sans.
```



```
blocked = results[results["Action"].str.contains("Block", na=False)]
```

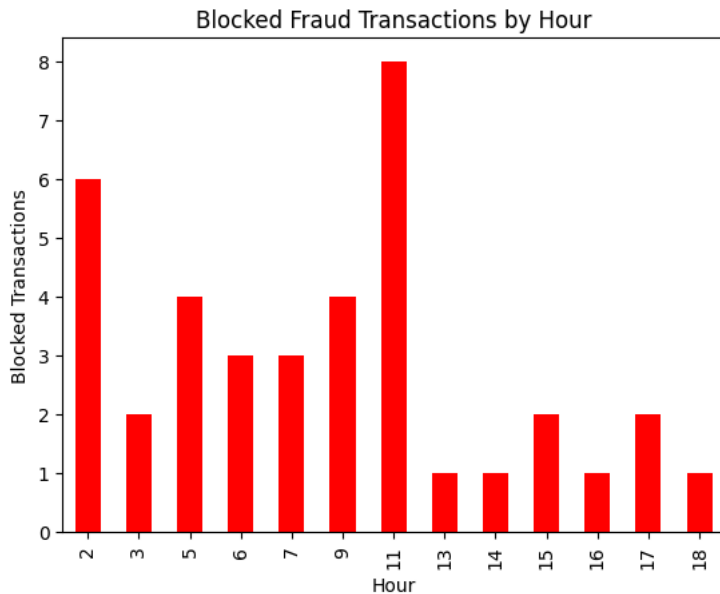
```
blocked.groupby("Hour").size().plot(kind="bar", color="red")
```

```
plt.title("Blocked Fraud Transactions by Hour")
```

```
plt.xlabel("Hour")
```

```
plt.ylabel("Blocked Transactions")
```

```
plt.show()
```



```
print("● Approved:", (results["Action"].str.contains("Approve")).sum())
print("● Manual Review:", (results["Action"].str.contains("Manual")).sum())
print("● Blocked:", (results["Action"].str.contains("Block")).sum())
print("Average Risk Score:", round(results["Risk_Score"].mean(), 2))
```

● Approved: 19313
 ● Manual Review: 4
 ● Blocked: 38
 Average Risk Score: 0.45

```
!pip install streamlit pyngrok
```

```
Collecting streamlit
  Downloading streamlit-1.63.0-py3-none-any.whl.metadata (10 kB)
Collecting pyngrok
  Downloading pyngrok-8.1.2-py3-none-any.whl.metadata (8.6 kB)
Requirement already satisfied: altair!=5.4.0,!5.4.1,<7,>=5.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (5.2.0)
Requirement already satisfied: click<9,>=7.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (8.5.0)
Requirement already satisfied: numpy<3,>=1.23 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.1.3)
Requirement already satisfied: packaging>=20 in /usr/local/lib/python3.13/dist-packages (from streamlit) (26.3)
Requirement already satisfied: pandas<4,>=1.4.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.3)
Requirement already satisfied: pillow<13,>=7.1.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (11.3.0)
Collecting pydeck<1,>=0.8.0b4 (from streamlit)
  Downloading pydeck-0.9.3-py2.py3-none-any.whl.metadata (4.2 kB)
Requirement already satisfied: protobuf<8,>=5.26.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (5.29.6)
Requirement already satisfied: pyarrow!=25.0.0,<26,>=7.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (18.1.0)
Requirement already satisfied: requests<3,>=2.27 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.32.4)
Requirement already satisfied: toml<2,>=0.10.1 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.10.2)
Requirement already satisfied: typing-extensions<5,>=4.10.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.16.0)
Requirement already satisfied: starlette<2,>=0.46.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (1.6.0)
Requirement already satisfied: uvicorn<1,>=0.30.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.52.4)
Requirement already satisfied: httpx<1,>=0.6.3 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.8.0)
Requirement already satisfied: anyio<5,>=4.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (4.14.2)
Requirement already satisfied: python-multipart<1,>=0.0.10 in /usr/local/lib/python3.13/dist-packages (from streamlit) (0.0.32)
Requirement already satisfied: websockets<17,>=12.0.0 in /usr/local/lib/python3.13/dist-packages (from streamlit) (15.0.1)
Requirement already satisfied: itsdangerous<3,>=2.1.2 in /usr/local/lib/python3.13/dist-packages (from streamlit) (2.2.0)
Requirement already satisfied: watchdog<7,>=2.1.5 in /usr/local/lib/python3.13/dist-packages (from streamlit) (6.0.0)
Requirement already satisfied: PyYAML>=5.1 in /usr/local/lib/python3.13/dist-packages (from pyngrok) (6.0.3)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (3.1.4)
Requirement already satisfied: jsonschema>=3.0 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (4.23.0)
Requirement already satisfied: narwhals>=1.14.2 in /usr/local/lib/python3.13/dist-packages (from altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (1.42.0)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.13/dist-packages (from anyio<5,>=4.0.0->streamlit) (3.19)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.4.0->streamlit) (2.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.4.0->streamlit) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas<4,>=1.4.0->streamlit) (2025.2)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->streamlit) (3.4.0)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->streamlit) (2.3.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.13/dist-packages (from requests<3,>=2.27->streamlit) (2025.8.3)
Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.13/dist-packages (from uvicorn<1,>=0.30.0->streamlit) (0.16.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.13/dist-packages (from jinja2->altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (3.0.2)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (25.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (0.36.0)
Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.13/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<7,>=5.0.0->streamlit) (0.25.0)
```

```
import streamlit as st
import pandas as pd

st.set_page_config(page_title="Razorpay AI Risk Dashboard", layout="wide")

# Load CSV
results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")

st.title("🚧 Razorpay AI Fraud Detection Dashboard")
st.caption("AI-powered payment risk scoring for merchants")

# KPI Cards
approved = (results["Action"].str.contains("Approve")).sum()
review = (results["Action"].str.contains("Manual")).sum()
blocked = (results["Action"].str.contains("Block")).sum()
avg_risk = round(results["Risk_Score"].mean(),2)

c1,c2,c3,c4 = st.columns(4)
c1.metric("Approved", approved)
c2.metric("Manual Review", review)
c3.metric("Blocked", blocked)
c4.metric("Avg Risk Score", avg_risk)

st.divider()

st.subheader("Payment Decision Summary")
st.bar_chart(results["Action"].value_counts())

st.subheader("Blocked Transactions by Hour")
blocked_df = results[results["Action"].str.contains("Block", na=False)]
st.bar_chart(blocked_df.groupby("Hour").size())

st.subheader("High Risk Transactions")
st.dataframe(
    blocked_df[["Risk_Score", "Hour", "Scaled_Amount", "Actual_Class"]]
)
```

[illegible]

```

2026-09-05 16:18:37.892 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:37.894 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.186 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.187 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.188 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.189 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.190 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.191 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.216 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.217 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.220 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.221 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.222 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.223 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.231 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.232 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.234 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
2026-09-05 16:18:41.235 Thread 'MainThread': missing ScriptRunContext! This warning can be ignored when running in bare mode.
DeltaGenerator()

```

```
!pip install gradio pandas plotly
```

```

Requirement already satisfied: gradio in /usr/local/lib/python3.13/dist-packages (6.26.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.13/dist-packages (2.2.3)
Requirement already satisfied: plotly in /usr/local/lib/python3.13/dist-packages (5.24.1)
Requirement already satisfied: anyio<5.0,>=3.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (4.14.2)
Requirement already satisfied: audioop-lts<=0.2.2 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.2.2)
Requirement already satisfied: brotli>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (1.2.0)
Requirement already satisfied: fastapi<1.0,>=0.115.2 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.141.1)
Requirement already satisfied: gradio-client==2.6.1 in /usr/local/lib/python3.13/dist-packages (from gradio) (2.6.1)
Requirement already satisfied: groovy~=0.1 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.1.2)
Requirement already satisfied: hf-gradio<1.0,>=0.4.1 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.4.1)
Requirement already satisfied: httpx<1.0,>=0.24.1 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.28.1)
Requirement already satisfied: huggingface-hub<2.0,>=1.16.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (1.28.0)
Requirement already satisfied: jinja2<4.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (3.1.6)
Requirement already satisfied: markupsafe<4.0,>=2.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (3.0.3)
Requirement already satisfied: numpy<3.0,>=1.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (2.1.3)
Requirement already satisfied: orjson~=3.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (3.12.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from gradio) (26.3)
Requirement already satisfied: pillow<13.0,>=8.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (11.3.0)
Requirement already satisfied: pydantic<3.0,>=2.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (2.13.4)
Requirement already satisfied: pydub<1.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.25.1)
Requirement already satisfied: python-multipart<1.0,>=0.0.18 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.0.32)
Requirement already satisfied: pytz>=2017.2 in /usr/local/lib/python3.13/dist-packages (from gradio) (2025.2)
Requirement already satisfied: pyyaml<7.0,>=5.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (6.0.3)
Requirement already satisfied: safehttpx<0.2.0,>=0.1.7 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.1.7)
Requirement already satisfied: semantic-version~=2.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (2.10.0)
Requirement already satisfied: starlette<2.0,>=1.0.1 in /usr/local/lib/python3.13/dist-packages (from gradio) (1.6.0)
Requirement already satisfied: tomlkit<0.15.0,>=0.12.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.14.0)
Requirement already satisfied: typer<1.0,>=0.12 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.27.1)
Requirement already satisfied: typing-extensions~=4.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (4.16.0)
Requirement already satisfied: uvicorn>=0.14.0 in /usr/local/lib/python3.13/dist-packages (from gradio) (0.52.4)
Requirement already satisfied: fsspec in /usr/local/lib/python3.13/dist-packages (from gradio-client==2.6.1->gradio) (2025.3.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.13/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.13/dist-packages (from pandas) (2026.3)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.13/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: idna>=2.8 in /usr/local/lib/python3.13/dist-packages (from anyio<5.0,>=3.0->gradio) (3.19)
Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.13/dist-packages (from fastapi<1.0,>=0.115.2->gradio) (0.0.2)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.13/dist-packages (from fastapi<1.0,>=0.115.2->gradio) (0.0.2)
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-packages (from httpx<1.0,>=0.24.1->gradio) (2026.7.22)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1.0,>=0.24.1->gradio) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1.0,>=0.24.1->gradio) (0.16.0)
Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0,>=1.16.0->gradio) (8.4.2)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0,>=1.16.0->gradio) (3.10.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.5.2 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0,>=1.16.0->gradio) (1.5.2)
Requirement already satisfied: tqdm>=4.42.1 in /usr/local/lib/python3.13/dist-packages (from huggingface-hub<2.0,>=1.16.0->gradio) (4.42.1)
Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0,>=2.0->gradio) (0.6.0)
Requirement already satisfied: pydantic-core==2.46.4 in /usr/local/lib/python3.13/dist-packages (from pydantic<3.0,>=2.0->gradio) (2.46.4)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.13/dist-packages (from typer<1.0,>=0.12->gradio) (1.5.4)
Requirement already satisfied: rich>=13.8.0 in /usr/local/lib/python3.13/dist-packages (from typer<1.0,>=0.12->gradio) (13.9.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer<1.0,>=0.12->gradio) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.13/dist-packages (from rich>=13.8.0->typer<1.0,>=0.12->gradio) (2.18.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.13/dist-packages (from markdown-it-py>=2.2.0->rich>=13.8.0->typer<1.0,>=0.12->gradio) (0.1.2)

```

```

import gradio as gr
import pandas as pd
import plotly.express as px

```

```
results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")
```

```
approved = results["Action"].str.contains("Approve").sum()
review = results["Action"].str.contains("Manual").sum()
blocked = results["Action"].str.contains("Block").sum()

fig = px.bar(results.groupby("Hour").size().reset_index(),
             x="Hour", y=0, title="Transactions by Hour")

with gr.Blocks(title="Razorpay AI Risk Dashboard") as demo:
    gr.Markdown("# 🚧 Razorpay AI Risk Dashboard")

    with gr.Row():
        gr.Textbox(value=str(approved), label="🟢 Approved")
        gr.Textbox(value=str(review), label="🟡 Manual Review")
        gr.Textbox(value=str(blocked), label="🔴 Blocked")

    gr.Plot(fig)
    gr.Dataframe(results.head(20), label="Top Transactions")

demo.launch(share=True)
```

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: <https://f056a0978b6cc4fff1f.gradio.live>

This share link is temporary and will last for up to 1 week (best effort). For free permanent hosting and GPU upgrades, run `gra



No interface is running right now

```
%%writefile app.py
import gradio as gr
import pandas as pd
import plotly.express as px

results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")

approved = (results["Action"]=="Approve Transaction").sum()
review = (results["Action"]=="Manual Review").sum()
blocked = results["Action"].str.contains("Block").sum()

fig = px.bar(results.groupby("Hour").size().reset_index(),
             x="Hour", y=0, title="Transactions by Hour")

with gr.Blocks(title="Razorpay AI Risk Dashboard") as demo:
    gr.Markdown("# 📊 Razorpay AI Risk Dashboard")
    with gr.Row():
        gr.Textbox(value=str(approved), label="🟢 Approved")
        gr.Textbox(value=str(review), label="🟡 Manual Review")
        gr.Textbox(value=str(blocked), label="🔴 Blocked")
    gr.Plot(fig)
    gr.Dataframe(results.head(20), label="Top Transactions")

demo.launch()
```

Overwriting app.py

```
from google.colab import files
files.download("app.py")
```

```
import gradio as gr
import pandas as pd
import plotly.express as px

# Load results
results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")

# ---- FIX ACTION COLUMN ----
results["Action"] = results["Action"].astype(str).str.lower()

approved = results["Action"].str.contains("approve").sum()
review = results["Action"].str.contains("review").sum()
blocked = results["Action"].str.contains("block").sum()

# Hour chart
hourly = results.groupby("Hour").size().reset_index(name="Transactions")

fig = px.bar(
    hourly,
    x="Hour",
    y="Transactions",
    color="Transactions",
    color_continuous_scale="Blues",
    template="plotly_white",
    title="Transactions by Hour"
)

fig.update_layout(
    plot_bgcolor="#F5F7FB",
    paper_bgcolor="white",
    font=dict(color="#1A1F36"),
    title_font_size=22
)

with gr.Blocks(theme=gr.themes.Soft(primary_hue="blue")) as demo:
    gr.Markdown("# 📊 Razorpay AI Fraud Detection Dashboard")
    gr.Markdown("Real-time merchant transaction risk monitoring and fraud analytics.")

    with gr.Row():
        gr.Markdown(f"### 🟢 Approved\n# {approved}")
        gr.Markdown(f"### 🟡 Manual Review\n# {review}")
        gr.Markdown(f"### 🔴 Blocked\n# {blocked}")

    gr.Plot(fig)
```

```

gr.Markdown("## 🚨 High Risk / Blocked Transactions")
gr.DataFrame(
    results[results["Action"].str.contains("block")].head(20),
    label="Blocked Transactions"
)

demo.launch(share=True)

```

/tmp/ipykernel_2345/1729851886.py:35: UserWarning:

The parameters have been moved from the Blocks constructor to the launch() method in Gradio 6.0: theme. Please pass these parameters

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No interface is running right now

```

import pandas as pd
import plotly.express as px
import gradio as gr

# Load prediction results
results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")

# ----- FIX KPIs -----
approved = len(results[results["Action"].str.contains("Approve", case=False, na=False)])
review = len(results[results["Action"].str.contains("Review", case=False, na=False)])
blocked = len(results[results["Action"].str.contains("Block", case=False, na=False)])

# Hourly transactions
hourly = results.groupby("Hour").size().reset_index(name="Transactions")

# Blue Razorpay Chart
fig = px.bar(
    hourly,
    x="Hour",
    y="Transactions",
    color="Transactions",
    color_continuous_scale="Blues",
    title="Fraud Transactions by Hour"
)

fig.update_layout(
    template="plotly_white",
    paper_bgcolor="#F5F9FF",
    plot_bgcolor="#F5F9FF",
    title_x=0.5,
    title_font_size=22,
    font=dict(color="#072654", size=14),

```



```

        coloraxis_showscale=False
    )

css = """
.gradio-container{
    background:#F5F9FF;
    font-family:Arial;
}
"""

with gr.Blocks(theme=gr.themes.Soft(primary_hue="blue"), css=css) as demo:

    gr.Markdown("""
    # 🏡 Razorpay AI Fraud Detection Dashboard
    ### Real-Time Merchant Risk Intelligence Platform

    AI-powered payment fraud detection, merchant risk monitoring, and suspicious transaction analytics.
    """)

    # KPI Cards
    with gr.Row():
        gr.Markdown(f"""
        <div style='background:#E8F5E9;padding:18px;border-radius:14px;text-align:center'>
        <h3>🟢 Approved</h3>
        <h1 style='color:#16A34A'>{approved:,}</h1>
        </div>
        """)

        gr.Markdown(f"""
        <div style='background:#FFF8E1;padding:18px;border-radius:14px;text-align:center'>
        <h3>🟡 Manual Review</h3>
        <h1 style='color:#CA8A04'>{review}</h1>
        </div>
        """)

        gr.Markdown(f"""
        <div style='background:#FDECEA;padding:18px;border-radius:14px;text-align:center'>
        <h3>🔴 Blocked</h3>
        <h1 style='color:#DC2626'>{blocked}</h1>
        </div>
        """)

    gr.Markdown("## 📊 Transaction Volume Monitoring")
    gr.Plot(fig)

    gr.Markdown("## 🚨 High-Risk Transactions Flagged by AI")
    gr.DataFrame(
        results[results["Action"].str.contains("Block", case=False, na=False)].head(20),
        interactive=False
    )

    gr.Markdown(f"""
    ## 📋 Risk Summary

    - Total Transactions: {len(results):,}
    - Approved Transactions: {approved:,}
    - Manual Review Required: {review}
    - Blocked Fraud Transactions: {blocked}
    - Fraud Detection Model: Isolation Forest
    - Use Case: Razorpay Merchant Fraud Risk Monitoring
    """)

demo.launch(share=True)

```

/tmp/ipykernel_2345/3827189155.py:43: UserWarning:

The parameters have been moved from the Blocks constructor to the launch() method in Gradio 6.0: theme, css. Please pass these p

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://1173911bfff372d3b90.gradio.live>

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No interface is running right now

```
import gradio as gr
import pandas as pd
import plotly.express as px

# Load CSV
results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")

# Fix Action names
results["Action"] = results["Action"].astype(str).str.upper()

approved = (results["Action"] == "APPROVE").sum()
review = (results["Action"] == "REVIEW").sum()
blocked = results["Action"].str.contains("BLOCK").sum()

# Hour column
results["Hour"] = pd.to_datetime(results["Time"], unit='s').dt.hour # Changed 'Timestamp' to 'Time' and added unit='s' if 'Time

# Blue Razorpay chart
hourly = results.groupby("Hour").size().reset_index(name="Transactions")

fig = px.bar(
    hourly,
    x="Hour",
    y="Transactions",
    color="Transactions",
    color_continuous_scale="Blues",
    text="Transactions",
    title="🕒 Hourly Transaction Volume"
)

fig.update_layout(
    template="plotly_white",
    plot_bgcolor="#F8FAFC",
    paper_bgcolor="FFFFFF",
    font=dict(color="#072654", size=14),
    title_font=dict(size=24, color="#072654"),
    coloraxis_colorbar_title="Transactions"
)

fig.update_traces(
```

```

        marker_line_color="#0A2540",
        marker_line_width=1,
        textposition="outside"
    )

# Risk Heatmap
heatmap = px.density_heatmap(
    results,
    x="Hour",
    y="Risk_Score", # Changed 'Risk Score' to 'Risk_Score'
    color_continuous_scale="Turbo",
    title="🔥 Fraud Risk Heatmap"
)

# Donut Chart
pie = px.pie(
    names=["Approved", "Review", "Blocked"],
    values=[approved, review, blocked],
    hole=0.65,
    color_discrete_sequence=["#00C853", "#FFD600", "#FF1744"],
    title="Transaction Decision Distribution"
)

with gr.Blocks() as demo:

    gr.Markdown("""
    # 📊 Razorpay AI Fraud Detection & Merchant Risk Monitoring Dashboard

    ### Payment Risk Intelligence Platform
    Built by Riddhi Pathak | MA Economics (FinTech)
    """)

    with gr.Row():
        gr.Markdown(f"### 🟢 Approved\n# {approved:,}")
        gr.Markdown(f"### 🟡 Manual Review\n# {review:,}")
        gr.Markdown(f"### 🔴 Blocked\n# {blocked:,}")

    gr.Markdown("---")
    gr.Plot(fig)

    with gr.Row():
        gr.Plot(pie)
        gr.Plot(heatmap)

    gr.Markdown("## 📄 Top Fraud Transactions")
    gr.Dataframe(results.head(20))

demo.launch(share=True, inline=True, theme=gr.themes.Soft(primary_hue="blue"))

```

/tmp/ipykernel_2345/1293833332.py:64: UserWarning:

The parameters have been moved from the Blocks constructor to the launch() method in Gradio 6.0: theme. Please pass these parameters

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This share link is temporary and will last for up to 1 week (best effort). For free permanent hosting and GPU upgrades, run `gradio



No interface is running right now

```
import pandas as pd
import plotly.express as px
import gradio as gr

# Load prediction results
results = pd.read_csv("Razorpay_AI_Risk_Fraud_Model_Riddhi_Results.csv")

# ----- FIX KPIs -----
approved = len(results[results["Action"].str.contains("Approve", case=False, na=False)])
review = len(results[results["Action"].str.contains("Review", case=False, na=False)])
blocked = len(results[results["Action"].str.contains("Block", case=False, na=False)])

# Hourly transactions
hourly = results.groupby("Hour").size().reset_index(name="Transactions")

# Blue Razorpay Chart
fig = px.bar(
    hourly,
    x="Hour",
    y="Transactions",
    color="Transactions",
    color_continuous_scale="Blues",
    title="Fraud Transactions by Hour"
)

fig.update_layout(
    template="plotly_white",
    paper_bgcolor="#F5F9FF",
    plot_bgcolor="#F5F9FF",
    title_x=0.5,
    title_font_size=22,
    font=dict(color="#072654", size=14),
    coloraxis_showscale=False
)

css = """
.gradio-container{
    background:#F5F9FF;
    font-family:Arial;
}
```

```

"""

with gr.Blocks(theme=gr.themes.Soft(primary_hue="blue"), css=css) as demo:

    gr.Markdown("""
    # 🏪 Razorpay AI Fraud Detection Dashboard
    ### Real-Time Merchant Risk Intelligence Platform

    AI-powered payment fraud detection, merchant risk monitoring, and suspicious transaction analytics.
    """)

    # KPI Cards
    with gr.Row():
        gr.Markdown(f"""
        <div style='background:#E8F5E9;padding:18px;border-radius:14px;text-align:center'>
        <h3>🟢 Approved</h3>
        <h1 style='color:#16A34A'>{approved:,}</h1>
        </div>
        """)

        gr.Markdown(f"""
        <div style='background:#FFF8E1;padding:18px;border-radius:14px;text-align:center'>
        <h3>🟡 Manual Review</h3>
        <h1 style='color:#CA8A04'>{review}</h1>
        </div>
        """)

        gr.Markdown(f"""
        <div style='background:#FDECEA;padding:18px;border-radius:14px;text-align:center'>
        <h3>🔴 Blocked</h3>
        <h1 style='color:#DC2626'>{blocked}</h1>
        </div>
        """)

    gr.Markdown("### 📊 Transaction Volume Monitoring")
    gr.Plot(fig)

    gr.Markdown("### 🚨 High-Risk Transactions Flagged by AI")
    gr.DataFrame(
        results[results["Action"].str.contains("Block", case=False, na=False)].head(20),
        interactive=False
    )

    gr.Markdown(f"""
    ## 📋 Risk Summary

    - Total Transactions: {len(results):,}
    - Approved Transactions: {approved:,}
    - Manual Review Required: {review}
    - Blocked Fraud Transactions: {blocked}
    - Fraud Detection Model: Isolation Forest
    - Use Case: Razorpay Merchant Fraud Risk Monitoring
    """)

demo.launch(share=True)

```

```
/tmp/ipykernel_2345/3827189155.py:43: UserWarning:
```

The parameters have been moved from the Blocks constructor to the launch() method in Gradio 6.0: theme, css. Please pass these p

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()

* Running on public URL: <https://8e835ca5aaaa559bfc.gradio.live>

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Double-click (or enter) to edit

 Razorpay AI Fraud Detection & Merchant Risk Monitoring Dashboard

▼ MA Economics (FinTech) Portfolio Project

Author: Riddhi Pathak

Tools: Python • Pandas • Plotly • Gradio • Isolation Forest Business Problem

Payment gateways process millions of transactions daily.

This dashboard helps:

- Detect fraudulent payments using AI.
- Flag risky transactions for manual review.
- Block suspicious transactions.
- Monitor fraud trends in real time.

```
results.info()  
results.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19355 entries, 0 to 19354
Data columns (total 37 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Time                  19355 non-null  int64
1   V1                    19355 non-null  float64
2   V2                    19355 non-null  float64
3   V3                    19355 non-null  float64
4   V4                    19355 non-null  float64
5   V5                    19355 non-null  float64
6   V6                    19355 non-null  float64
7   V7                    19355 non-null  float64
8   V8                    19355 non-null  float64
9   V9                    19355 non-null  float64
10  V10                   19355 non-null  float64
11  V11                   19355 non-null  float64
12  V12                   19355 non-null  float64
13  V13                   19355 non-null  float64
14  V14                   19355 non-null  float64
15  V15                   19355 non-null  float64
16  V16                   19355 non-null  float64
17  V17                   19355 non-null  float64
18  V18                   19355 non-null  float64
19  V19                   19355 non-null  float64
20  V20                   19355 non-null  float64
21  V21                   19355 non-null  float64
22  V22                   19355 non-null  float64
23  V23                   19355 non-null  float64
24  V24                   19355 non-null  float64
25  V25                   19355 non-null  float64
26  V26                   19355 non-null  float64
27  V27                   19355 non-null  float64
28  V28                   19355 non-null  float64
29  Hour                  19355 non-null  int64
30  Scaled_Amount         19355 non-null  float64
31  Actual_Class          19355 non-null  int64
32  Predicted_Class       19355 non-null  int64
33  Fraud_Probability     19355 non-null  float64
34  Risk_Score            19355 non-null  float64
35  Risk_Level            19355 non-null  object
36  Action                19355 non-null  object
dtypes: float64(31), int64(4), object(2)
memory usage: 5.5+ MB
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8
count	19355.000000	19355.000000	19355.000000	19355.000000	19355.000000	19355.000000	19355.000000	19355.000000	19355.000000
mean	41637.001447	-0.260039	-0.036700	0.668189	0.174320	-0.273691	0.091583	-0.105717	0.059600
std	16642.906254	1.887221	1.662406	1.397204	1.361889	1.417684	1.317528	1.271956	1.159400
min	7.000000	-34.148234	-38.694338	-33.680984	-4.934328	-42.147898	-16.671588	-24.419483	-31.464134
25%	33232.500000	-1.024894	-0.594404	0.176237	-0.712462	-0.877693	-0.653604	-0.591919	-0.140170
50%	43459.000000	-0.255179	0.082109	0.747727	0.195226	-0.308661	-0.161252	-0.064970	0.072950
75%	54648.000000	1.154033	0.725845	1.372237	1.043638	0.245721	0.487490	0.420429	0.357450

Data Preprocessing Pipeline
max 80086.000000
CSV → Missing Values → Duplicate Removal → Feature Scaling → Isolation Forest Model AI Fraud Detection Model

Model Used: Isolation Forest

Purpose:

- Detect anomalies.
- Generate risk scores.
- Approve / Review / Block transactions.

```
pie = px.pie(
    names=["Approved", "Manual Review", "Blocked"],
    values=[approved, review, blocked],
    hole=0.6
)
pie.show()
```

